Stratus Weather Server

Professional Weather Station Management Software

Stratus is a professional desktop and server application for weather station management, data collection, and real-time monitoring. Built with modern web technologies, Stratus provides comprehensive weather data visualization and analysis capabilities.

> **CLOUD DEPLOYMENT NOTE** Stratus is designed for cloud deployment on Railway or similar platforms. All connections use TCP/IP - serial/RS232 connections are not available.

Features & Capabilities

Weather Station Integration

- **Multi-Protocol Support** - PakBus, HTTP/REST, MQTT, Modbus
- **Multi-Connection Types** - TCP/IP, WiFi, Cellular (4G/LTE), LoRaWAN, Satellite
- **Data Collection** - Scheduled and on-demand data collection from stations
- **Real-Time Monitoring** - Live data updates with configurable refresh intervals
- **Multi-Station Support** - Manage multiple stations from a single dashboard

Supported Hardware

Campbell Scientific Dataloggers:

- CR1000X, CR1000, CR6, CR3000
- CR800, CR850, CR300
- CR200X series

Other Platforms:

- Arduino IoT Cloud compatible devices

- ESP32/ESP8266 WiFi stations
- Davis Vantage series
- Generic HTTP/MQTT stations
- Custom industrial sensors via Modbus

Communication Options

- **TCP/IP** - Ethernet, WiFi via NL121 or similar
- **Cellular** - 4G/LTE modems with TCP gateway
- **LoRa/LoRaWAN** - Long-range IoT connectivity via network server
- **Satellite** - Remote area connectivity via IP gateway
- **HTTP API** - Direct REST API integration

Real-Time Monitoring Dashboard

Live Weather Display

- Real-time weather data visualization with auto-refresh
- Connection health monitoring
- Data validation with automatic quality control checks
- Configurable alerts & alarms for data thresholds

Measured Parameters

Atmospheric:

- Temperature (°C/°F)
- Relative Humidity (%)
- Barometric Pressure (hPa/mbar)
- Dew Point (°C/°F)
- Air Density (kg/m^3)

Wind:

- Wind Speed (km/h, m/s, mph, knots)
- Wind Direction (degrees, compass)

- Wind Gust (instantaneous maximum)
- Wind Power Density (W/m^2)

Solar:

- Solar Radiation (W/m^2)
- UV Index
- Photosynthetically Active Radiation (PAR)
- Sunshine Duration

Precipitation:

- Rainfall (mm/inch)
- Rain Rate (mm/hr)
- Daily/Monthly/Yearly Totals

Soil (Agriculture):

- Soil Temperature at multiple depths
- Soil Moisture (volumetric water content)
- Leaf Wetness

Air Quality:

- PM2.5 and PM10 particulate matter
- CO2 concentration
- Atmospheric visibility

System:

- Battery voltage and charge status
- Solar panel voltage
- Logger internal temperature

Solar Position Tracking

Real-time solar position calculated from station coordinates:

- **Sun Elevation** - Degrees above/below horizon (-90° to +90°)

- **Sun Azimuth** - Degrees from north (0° to 360°)
- **Nautical Dawn/Dusk** - When sun is 12° below horizon
- **Sunrise/Sunset** - Actual rise and set times
- **Solar Noon** - Time of maximum sun elevation
- **Day Length** - Hours of daylight

Calculations use NOAA solar position algorithms with millisecond precision.

Air Density Calculation

Real-time air density calculated from:

- Temperature (°C)
- Barometric Pressure (hPa)
- Relative Humidity (%)

Uses the ideal gas law with humidity correction. Standard reference: 1.225 kg/m^3 at sea level, 15°C.

Barometric Pressure

Dual display showing:

1. **Station Pressure** - Raw pressure at station altitude (hPa)
2. **Sea Level Pressure (QNH)** - Pressure calibrated to mean sea level (hPa)

The sea level correction uses the hypsometric equation accounting for station altitude and temperature.

Reference Evapotranspiration (ETo)

Calculated using the **FAO Penman-Monteith method** (FAO-56 standard), the internationally recognized standard for agricultural water management:

Required Inputs:

- Air temperature
- Relative humidity
- Wind speed at 2m height
- Solar radiation
- Station altitude
- Station latitude

Outputs:

- Instantaneous ETo rate (mm/hr)
- Daily ETo (mm/day)
- Weekly cumulative ETo (mm)
- Monthly cumulative ETo (mm)

Essential for irrigation scheduling, crop water requirements, and agricultural planning.

Battery Monitoring

Comprehensive battery status monitoring:

- Current voltage display (V)
- Visual status indicator (Critical/Low/Fair/Good/Excellent)
- Charge percentage estimation
- 24-hour voltage history chart

Voltage Thresholds (12V systems):

- 🔴 Critical: < 11.5V
- 🟠 Low: 11.5V - 12.0V
- 🟢 Good: 12.0V - 13.5V
- ⚡ Charging: > 13.5V

Wind Analysis (WMO/Beaufort Scale)

Wind Rose Charts

Traditional wind direction frequency distribution showing:

- **Wind Rose (Last 60 min)** - Recent wind patterns
- **Wind Rose (Today)** - Daily wind direction distribution
- **Wind Rose (Yesterday)** - Previous day comparison

Each rose displays direction frequency with speed-coded segments using official WMO/Beaufort scale colors.

Wind Speed Scatter Plots

Individual wind speed observations plotted on polar coordinates:

- **Wind Speed (Last 30 min)** - Recent observations
- **Wind Speed (Today)** - All today's observations
- **Wind Speed (Yesterday)** - Previous day observations

Features:

- Points plotted by direction (angle) and speed (radius)
- Color-coded by WMO/Beaufort scale:

Beaufort	Description	km/h	Color
0	Calm	< 1	Light Blue
1	Light Air	1-6	Sky Blue
2	Light Breeze	6-12	Blue
3	Gentle Breeze	12-20	Cyan
4	Moderate	20-29	Deep Blue
5	Fresh	29-39	Green
6	Strong	39-50	Yellow-Green
7	Near Gale	50-62	Yellow

8	Gale	62-75	Orange
9	Strong Gale	75-89	Red
10	Storm	89-103	Dark Red
11	Violent Storm	103-118	Maroon
12	Hurricane	> 118	Black-Red

- Statistics: Average, Max, Min speed, Dominant direction
- Interactive tooltips with exact values

Wind Compass

Real-time wind direction and speed display with animated compass rose and direction arrow.

Wind Power Analysis

Wind energy potential calculations for renewable energy assessment:

- Wind Power Density (W/m^2)
- Available wind energy based on air density and wind speed cubed

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Fire Danger Index

McArthur Forest Fire Danger Index (FFDI) calculation for fire weather monitoring:

Input Parameters:

- Temperature (°C)
- Relative Humidity (%)
- Wind Speed (km/h)
- Drought Factor

Danger Categories:

- 🟢 Low: 0-12

- 🟡 Moderate: 12-25
- 🟠 High: 25-50
- 🔴 Very High: 50-75
- ⚫ Extreme: > 75

Data Management

Local Database

- **SQLite Storage** - All data stored locally for complete offline access
- **Data Export** - Export to CSV, JSON formats
- **Historical Charts** - Interactive charts with zoom and pan
- **Data Retention** - Configurable retention policies

Dashboard Export

Export the complete dashboard as a professional multi-page PDF report:

1. Click the **Export** button in the dashboard header
2. Select **Save as PDF**
3. The system will:
    - Capture the full dashboard with print-optimized styling
    - Split content across multiple A4 pages
    - Add station name, date, and page numbers
    - Generate a downloadable PDF file

Interactive Charts

Weather Time Series

- Temperature, humidity, pressure trends
- Wind speed and direction history
- Solar radiation curves

- Precipitation accumulation

Data Block Charts

Condensed multi-parameter display showing recent values with mini-sparklines.

Heat Maps

24-hour × 7-day matrix showing parameter variations by time of day and day of week.

Station Map

Interactive OpenStreetMap integration showing:

- Station location marker
- Coordinates display
- Altitude indicator
- Click-to-set location for new stations

Settings & Configuration

Display Units

Parameter	Options
Temperature	Celsius, Fahrenheit
Wind Speed	km/h, m/s, mph, knots
Pressure	hPa, mbar, inHg, mmHg
Precipitation	mm, inches
Solar Radiation	W/m^2, $MJ/m^2/day$

Dashboard Configuration

- Configurable auto-refresh interval (5s to 60min)
- Chart time range selection (1hr to 7 days)
- Enable/disable individual parameter cards
- Light and Dark theme support

Data Quality Settings

- Minimum/Maximum bounds checking
- Rate-of-change limits
- Stuck sensor detection
- Gap filling options

Deployment Options

Desktop Application

- Windows installer (EXE)
- Single-user local deployment
- No internet connection required
- All data stored locally

Server Deployment

- Docker container support
- Railway, Render, or self-hosted
- Multi-user web access
- Remote station monitoring

Remote Access

- Cloudflare Tunnel integration for secure public access
- Custom domain support
- Auto-restart Windows services for 24/7 reliability

- HTTPS encryption

Technical Specifications

Technology Stack

- **Frontend:** React, TypeScript, Tailwind CSS, Recharts
- **Backend:** Node.js, Express, SQLite
- **Desktop:** Electron for native Windows application
- **Maps:** Leaflet with OpenStreetMap tiles
- **PDF Export:** jsPDF with html2canvas

System Requirements

Desktop:

- Windows 10/11 (64-bit)
- 4GB RAM minimum
- 100MB disk space
- USB port for serial connections

Server:

- Node.js 18+
- 512MB RAM minimum
- SQLite or PostgreSQL database